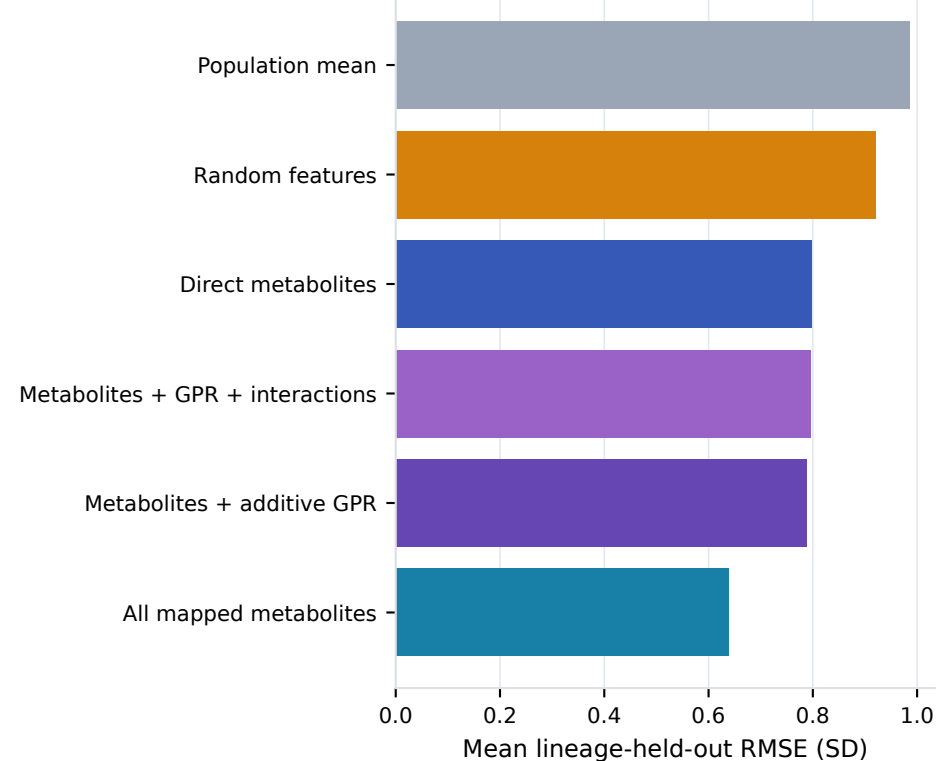
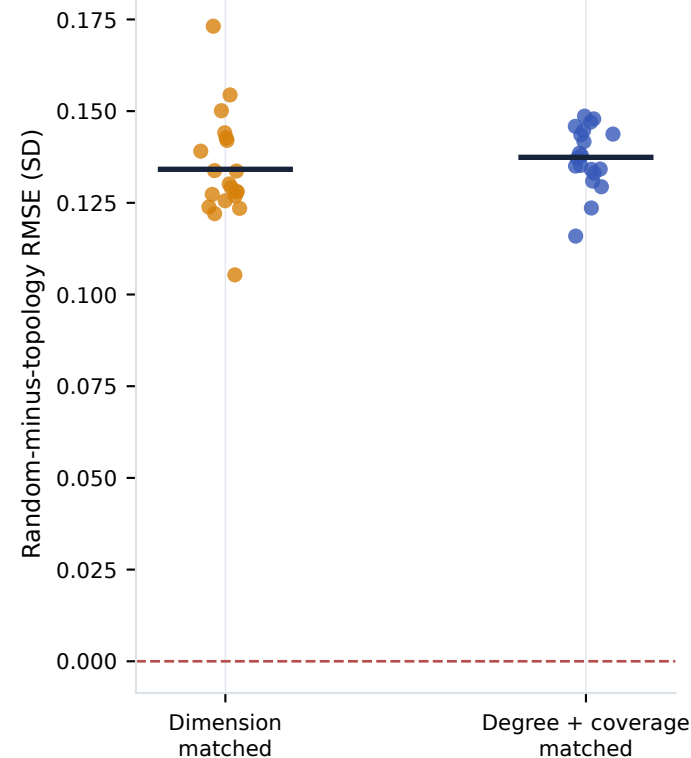
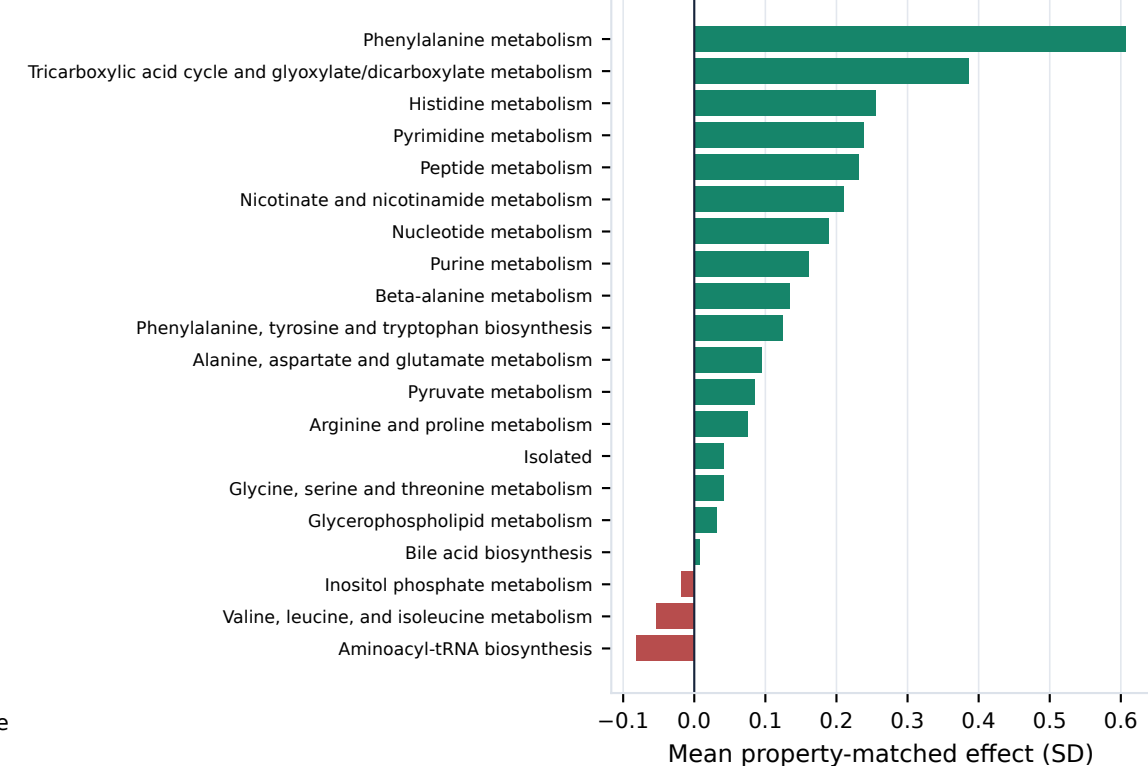
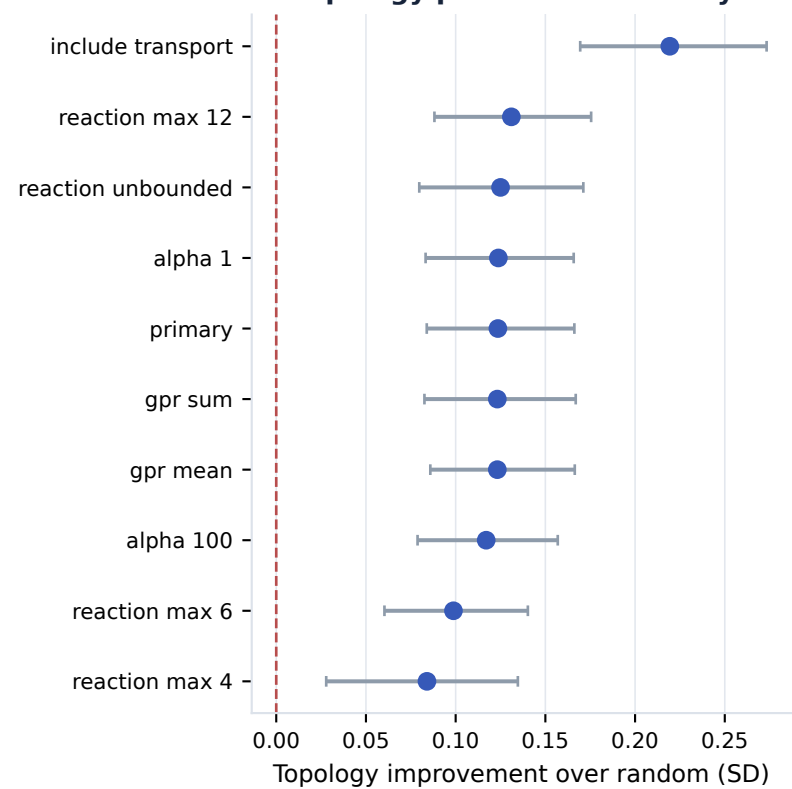
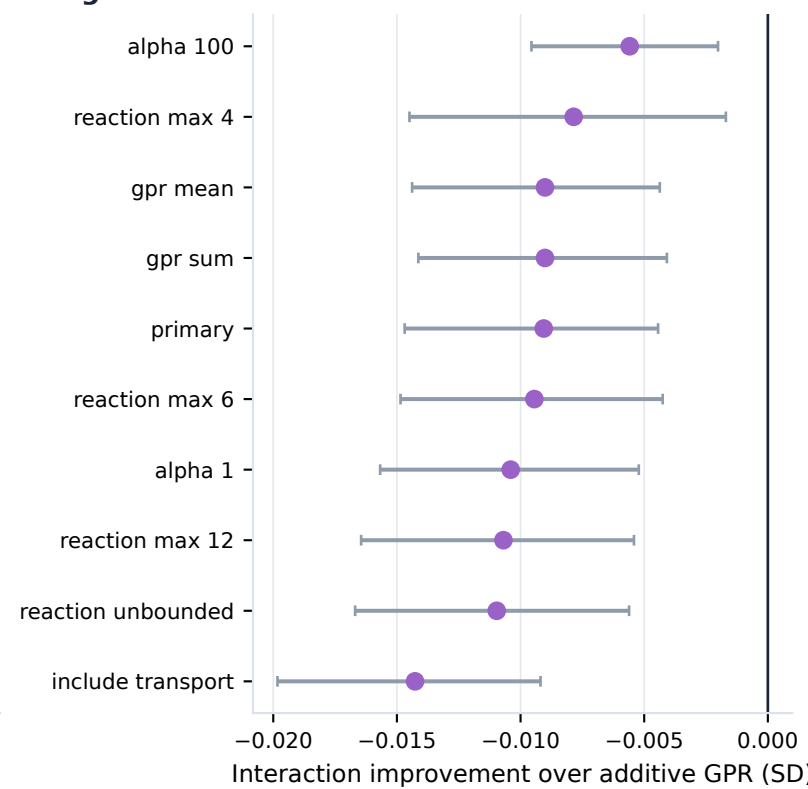


**A Reaction-neighborhood model ablation****B All 40 random-feature realizations favor topology****C Effects span 20 dominant HumanGEM subsystems****D Topology passes all ten analysis settings****E Interactions lose to additive GPR in every setting****F Mapping denominators bound generalization**